# Spatial Interpolation — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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# 1 Test Results — Spatial Interpolation (MKM-SP-001)

Tests run on **03 April 2026 at 20:15:57**.

Table 1: Test Results Summary — Spatial Interpolation (MKM-SP-001)

| | |
|---|---|
| Total Tests | 41 |
| Passed | 41 |
| Failed | 0 |
| Skipped | 0 |
| Duration | 0.01s |

Table 2: Detailed Test Results — Spatial Interpolation

| Test Description | Acceptance Criteria | Result | Time |
|---|---|---|---|
| Diagonal is one | — | Pass | 0.000s |
| Off diagonal between zero and one | `np.all(off_diag >= 0) and np.all(off_diag <= 1)` | Pass | 0.000s |
| Shape is n by n | `mat.shape == (n, n)` | Pass | 0.000s |
| Symmetric | — | Pass | 0.000s |
| Diagonal is one | — | Pass | 0.000s |
| Higher base correlation increases off diagonal | `np.all(off_high > off_low)` | Pass | 0.000s |
| Larger correlation_distance → slower exponential decay → higher off-diag. | `np.all(off_far >= off_near)` | Pass | 0.000s |
| Off diagonal positive | `np.all(off > 0)` | Pass | 0.000s |
| Returns square matrix | `mat.shape == (3, 3)` | Pass | 0.000s |
| Symmetric | — | Pass | 0.000s |
| Empty dataframe returns none | `build_spatial_index(df) is None` | Pass | 0.000s |
| None gauge data returns none | `build_spatial_index(None) is None` | Pass | 0.000s |
| With gauges returns kdtree | `isinstance(result, cKDTree)` | Pass | 0.000s |
| Lines 43-45: coordinates are converted to radians. | `tree.data[0][1] == pytest.approx(np.pi, rel=1e-6)` | Pass | 0.000s |
| Single gauge should still produce a valid KDTree. | `isinstance(tree, cKDTree); tree.n == 1` | Pass | 0.000s |
| Lines 42-46: non-empty DataFrame builds a cKDTree. | `isinstance(tree, cKDTree); tree.n == 3` | Pass | 0.000s |
| Equatorial degree approx 111km | `111_000 < d < 112_000` | Pass | 0.000s |
| London paris approx 340km | `330_000 < d < 350_000` | Pass | 0.000s |
| North south degree approx 111km | `110_000 < d < 112_000` | Pass | 0.000s |

| Test Description | Acceptance Criteria | Result | Time |
|---|---|---|---|
| Returns float | `isinstance(haversine_distance(51.0, -0.1, 52.0, 0.0), float)` | Pass | 0.000s |
| Same point is zero | `haversine_distance(51.5, -0.1, 51.5, -0.1) == pytest.appr...` | Pass | 0.001s |
| Symmetry | `d1 == pytest.approx(d2, rel=1e-9)` | Pass | 0.000s |
| All zero weights returns zero | `result == pytest.approx(7.5)` | Pass | 0.000s |
| Closer gauge has higher weight | `result < 5.0` | Pass | 0.000s |
| Empty list returns zero | `idw_interpolate_from_gauges([]) == 0.0` | Pass | 0.000s |
| Equal distance gauges averages wse | `result == pytest.approx(3.0)` | Pass | 0.000s |
| Near gauge returns wse directly | `result == pytest.approx(5.0)` | Pass | 0.000s |
| Power one changes result | `r2 != pytest.approx(r1, rel=0.01)` | Pass | 0.000s |
| Single far gauge | `result == pytest.approx(3.0)` | Pass | 0.000s |
| Line 158: power parameter affects weights. | `r3 < r1` | Pass | 0.000s |
| Line 149-150: empty input → 0.0. | `idw_interpolate_from_gauges([]) == 0.0` | Pass | 0.000s |
| Gauge at exactly min_distance=1.0 is NOT less than, so IDW used. | `result == pytest.approx(5.0)` | Pass | 0.000s |
| Multiple gauges produce weighted result. | `2.0 < result < 6.0; result < 4.0` | Pass | 0.000s |
| Line 156-157: dist ¡ min_distance → return wse. | `result == pytest.approx(42.0)` | Pass | 0.000s |
| At gauge location returns wse directly | `result == pytest.approx(3.0)` | Pass | 0.000s |
| Empty gauge data returns zero | `result == 0.0` | Pass | 0.000s |
| Idw returns weighted value | `2.5 <= result <= 4.5` | Pass | 0.000s |
| Single gauge | `result > 0` | Pass | 0.000s |
| Lines 119-120: distance ¡ 1m → direct return. | `result == pytest.approx(7.0)` | Pass | 0.000s |
| Line 97-98: no gauge data → 0.0. | `spatial_interpolate_wse(51.5, -0.1, {}, {}) == 0.0` | Pass | 0.000s |
| Lines 113-124: IDW with two equidistant gauges. | `result == pytest.approx(2.0, rel=0.01)` | Pass | 0.000s |

## 1.1 Test Document History

| Date | Version | Description | Author |
|---|---|---|---|
| 03-Apr-2026 | 1.0 | Initial test suite (41 tests) | D. Kelly |